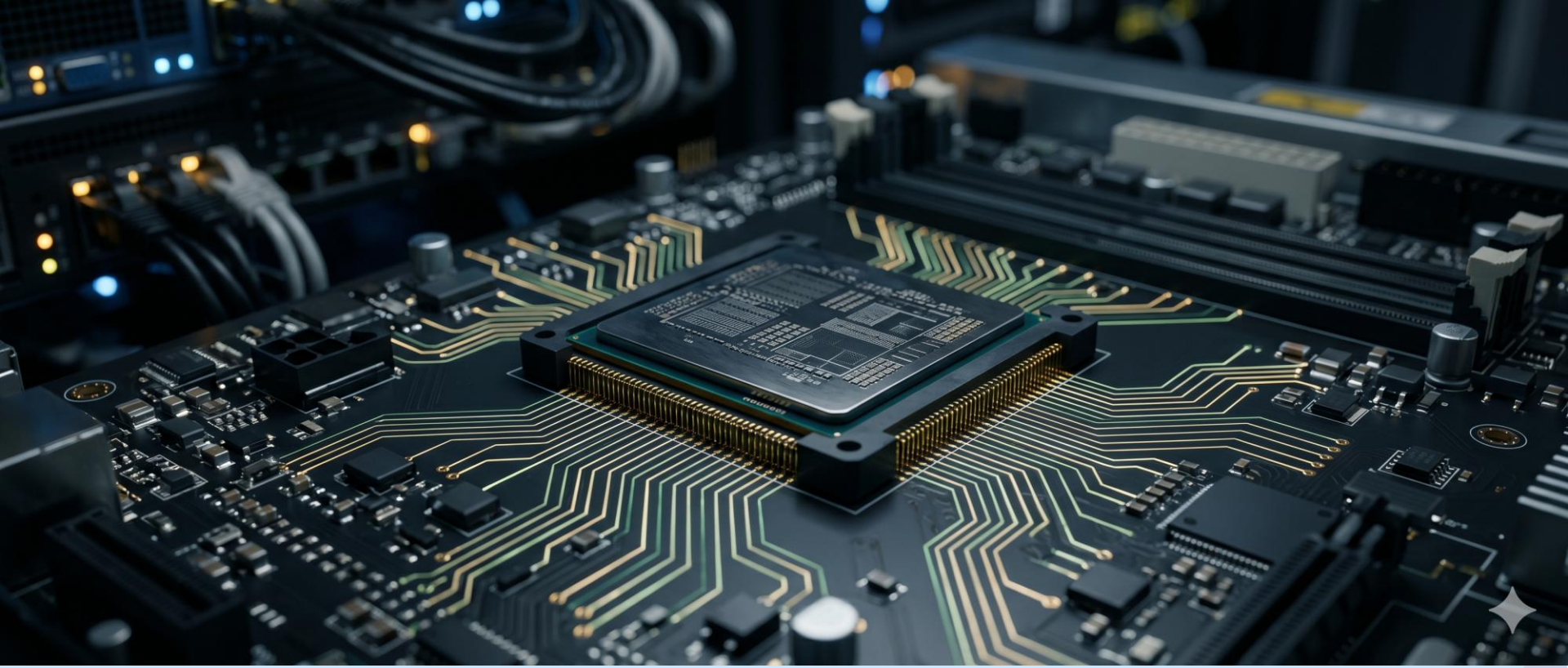


Beyond BMC: Architecting the Future of Servers

Chris Lin

May 2026

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Please note that while we have made every effort to list all significant risks, there may still be other risk factors that have not been identified or currently appear to be insignificant.

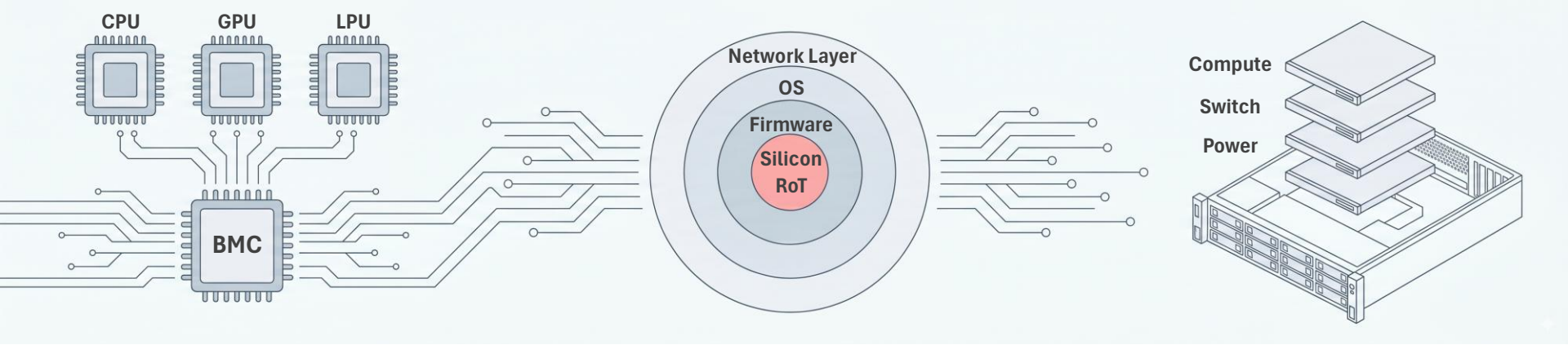
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請注意，儘管我們已盡力羅列各項重要風險，但仍可能存在尚未識別或當前看似不顯著的其他風險因素。

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Key Trends of Server Design



Heterogeneous Computing

- ✓ Increasing complexity
- ✓ XPU collaboration

Zero Trust & Root of Trust

- ✓ Platform RoT
- ✓ Silicon RoT
- ✓ PUF & PQC

Modular Design (Disaggregated)

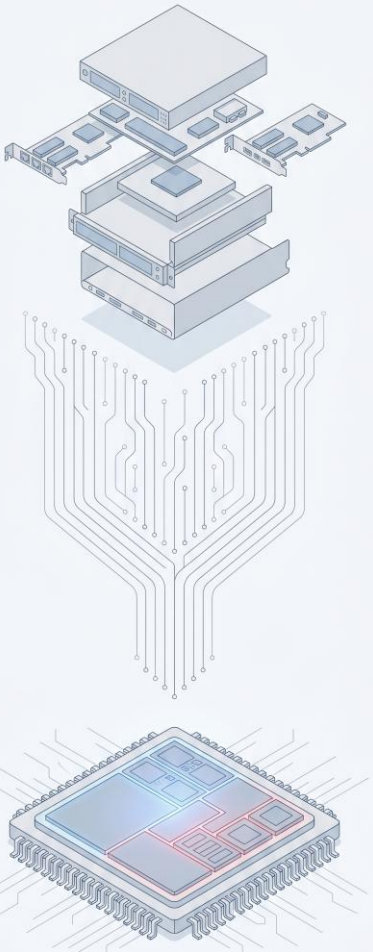
- ✓ Maintenance and upgrade
- ✓ Time-to-market
- ✓ Space efficiency
- ✓ Flexible manufacturing

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ASPEED Business Strategy

- Maximize *dollar content per server board with product suite;*
- Accelerating *product cadence.*

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Strategic Edges in Server Evolution

BMC: Setting the Standard

- ✓ Co-designed with hyperscalers and chip makers

SMC*: Multi-Card Architectures

- ✓ Empowering extreme scalability with efficient 2-level management

PRoT: Security for Platform

- ✓ Protection, detection, recovery

I/O Expander: Seamless Integration

- ✓ Bridging decoupled modules

*We further enhance security by integrating **Caliptra** into new generation of all chips.*

Note: Satellite Management Controller (SMC) is also known as Auxiliary Management Controller (AMC) or Bridge IC

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Ultimate Flexibility: Embedding FPGA into SoC

Partnership: Collaborating with FPGA leader **Lattice**.

Synergy: Embedding FPGA logic directly into AST1840 SoC

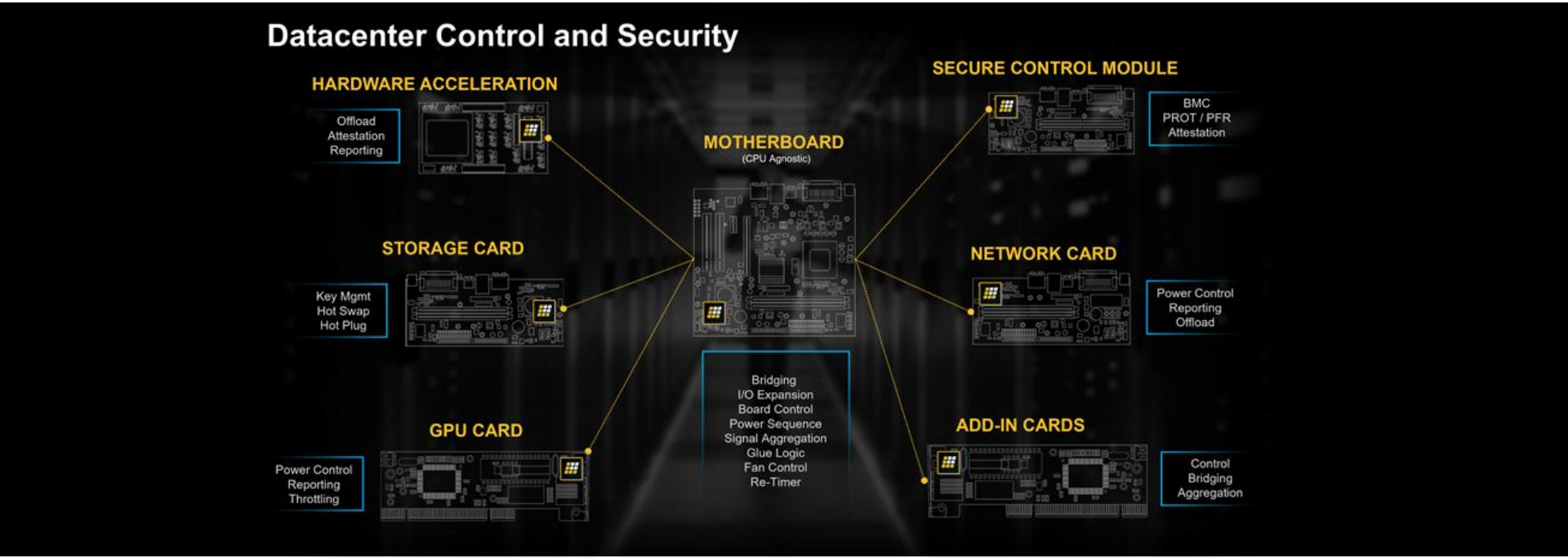
Flexibility: Delivering instant-on-programmable control for evolving server needs



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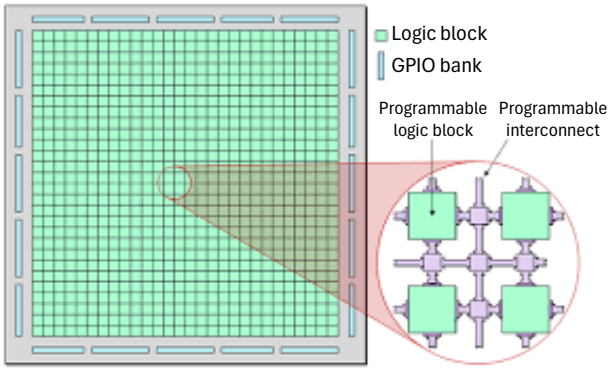
Lattice FPGA Applications

Lattice FPGA has multiple applications in datacenter and is working closely with ASPEED BMC.

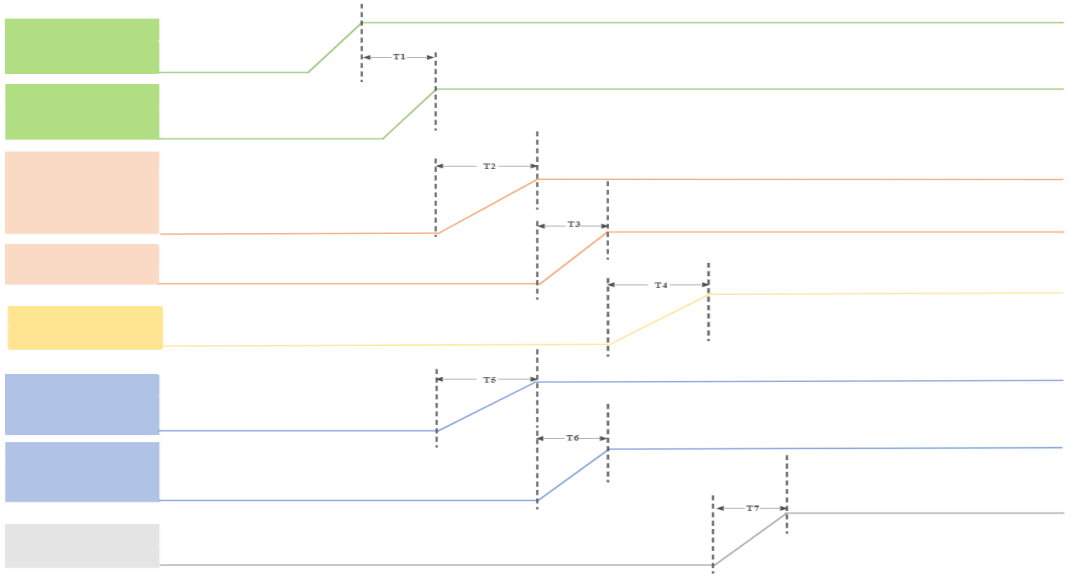


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FPGA Works as a Power Sequencer



Top view of FPGA



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The Lattice Advantages

Instant-On Control

✓ Zero-delay execution for precise power sequencing and system resets

Agile Customization

✓ Programmable logic to rapidly adapt to diverse server SKUs

Parallel Real-time Processing

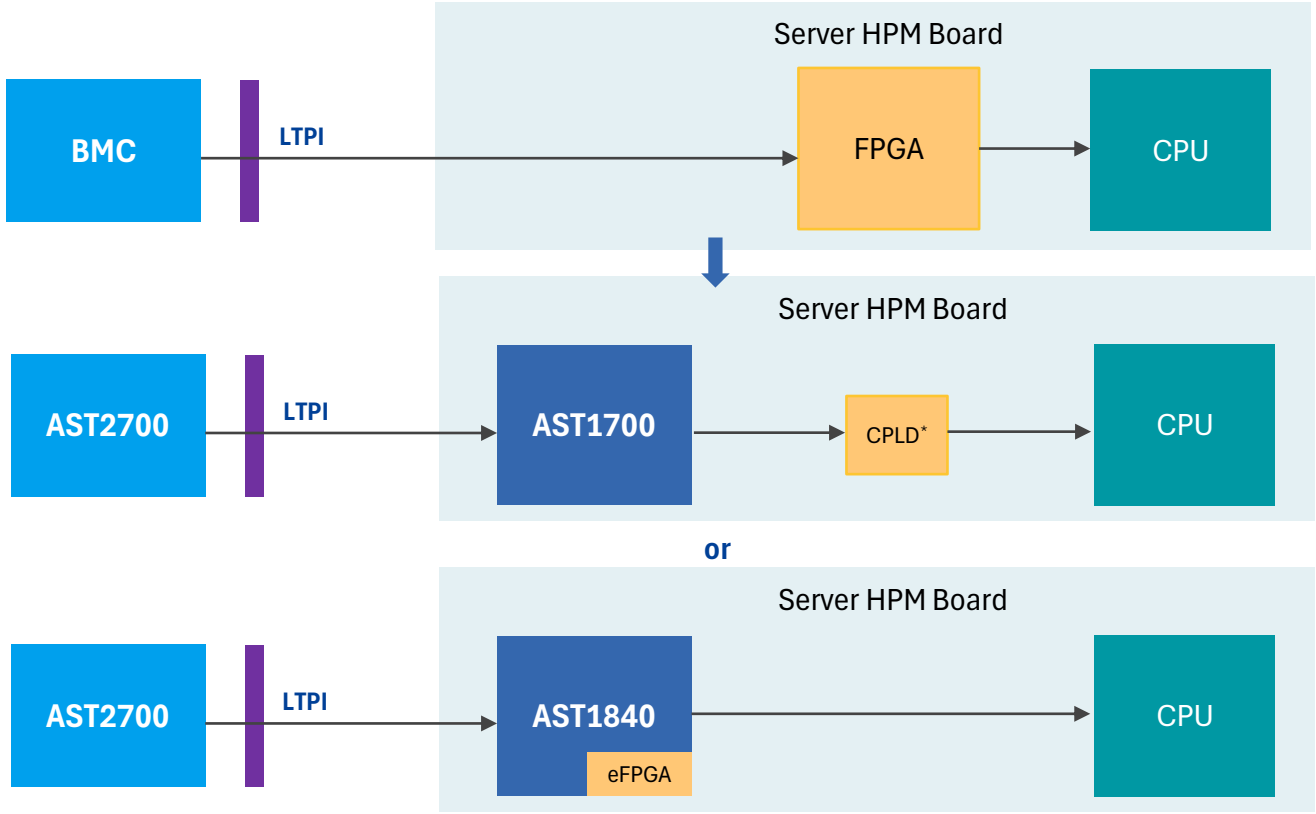
✓ Manage power, telemetry, and I/O tasks wit no firmware bottlenecks

I/O Consolidation & Bridging

✓ Unifying diverse interfaces (I2C, SPI) to simplify complex board designs

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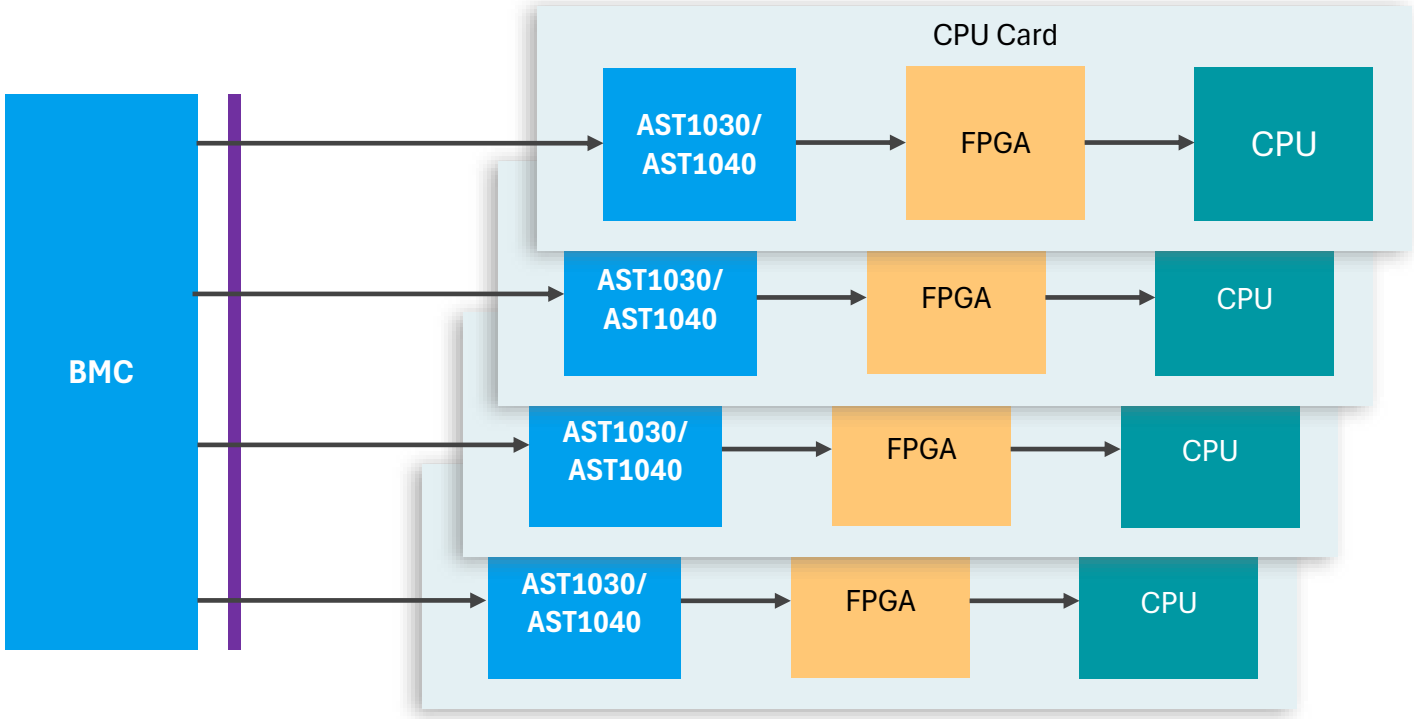
Evolving Server Architecture (Single Node)



Note: CPLD = small FPGA

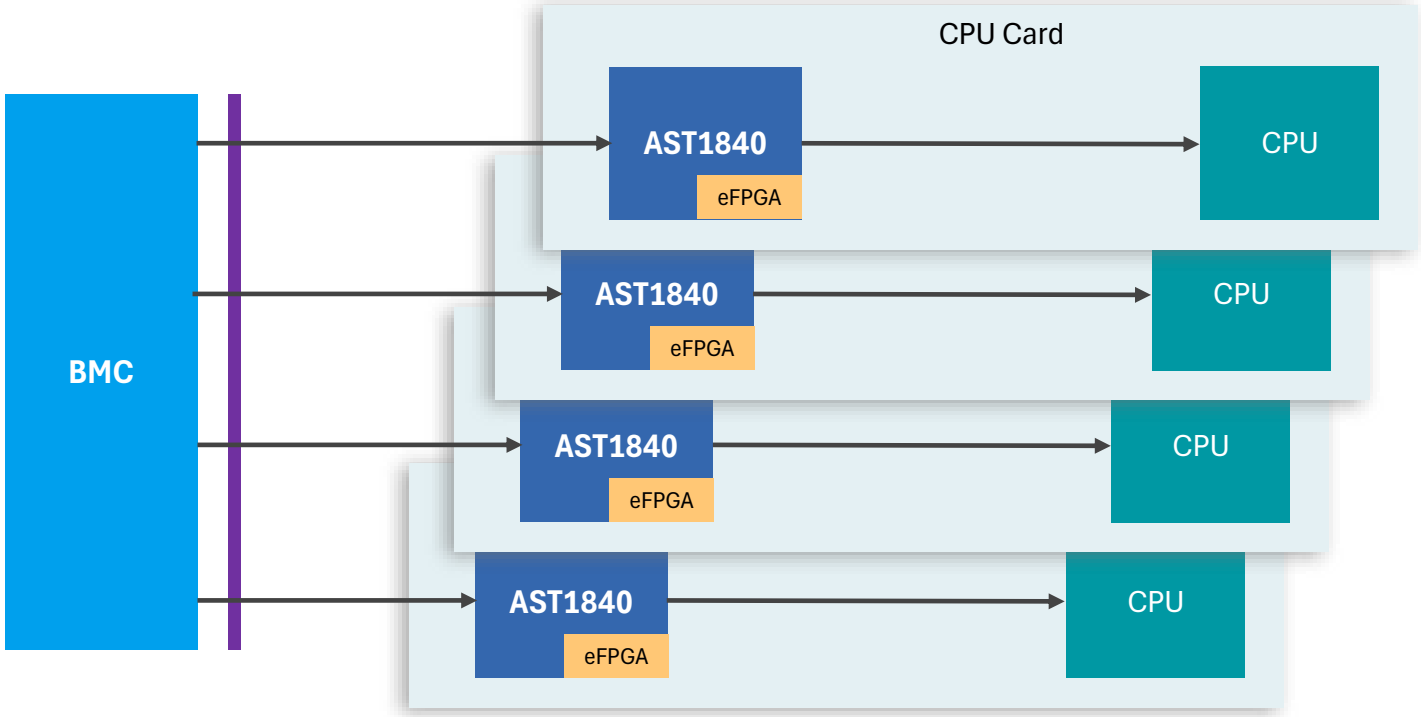
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Multi-Node with FPGA



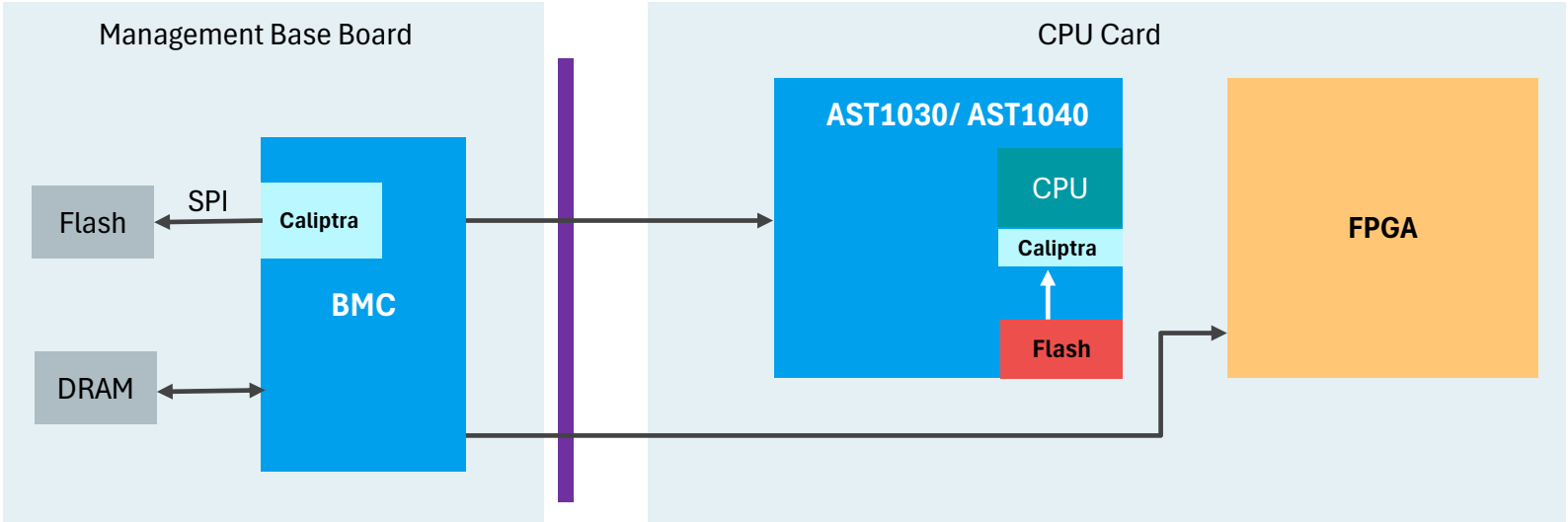
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Simplified Design with No FPGA



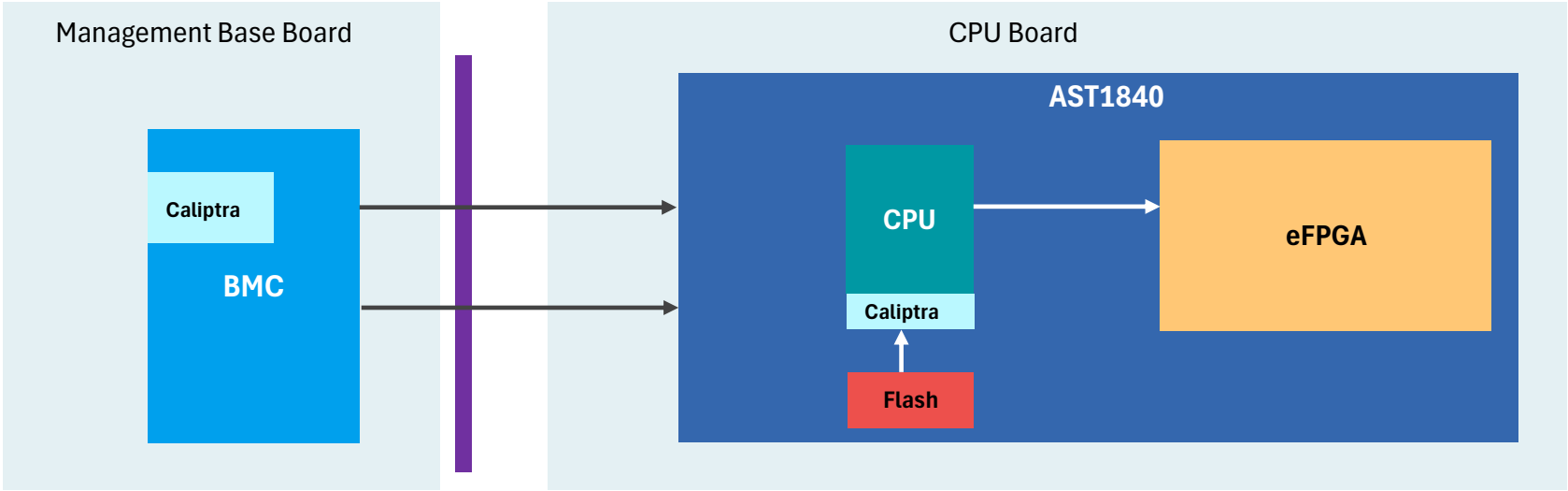
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Server with FPGA on CPU Board



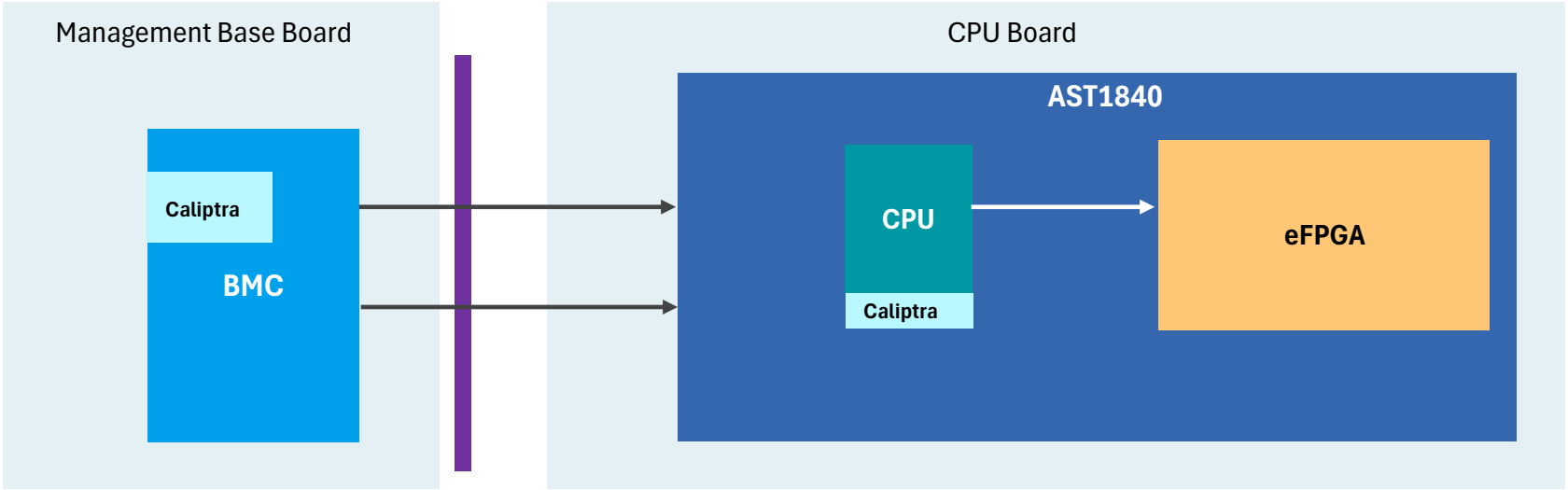
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No Streaming Boot with Flash



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Streaming Boot (No Flash)



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Core Value of AST1840

- ✓ **Enhanced Security:** *Sharing Caliptra with eFPGA*
- ✓ **Simplified Design:** *No external FPGA required*
- ✓ **Versatile:** *Suitable for various server designs*
- ✓ **Streaming Boot:** *Easy Update*

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Schedule and Potential TAM

Schedule: *Production-ready in 2027 Q3*

Potential TAM: *Multi million units*

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ASPEED 引領伺服器架構轉型

Beyond BMC

Driving the shift from single-node to multi-node heterogeneous computing.

The Best of Two Worlds

Maximizing product value with ASIC efficiency and eFPGA adaptability.

Streamlined and Agile

Simplifying design while ensuring hardware readiness for future upgrades.

Massive Potential

Expanding our TAM alongside next-gen server architectures.

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